



LOW-END AMD MOTHERBOARDS

MSI	Mercury	Krypton	Krypton
KM4M-L	KT400FDSX	C18G-400	NF2400U
VIA KM400	VIA KT400	Nforce2	Nforce2
Socket A	Socket A	Socket A	Socket A
Athlon XP 2800+	Athlon XP 2800+	Athlon XP 3200+	Athlon XP 3200+
333 / 2 / 2	333 / 3 / 3	333 / 2 / 3	333 / 3 / 3
Mini ATX	ATX	Mini-ATX	ATX
✓ / ✓ / 8X	✓ / No / 8X	✓ / ✓ / 8X	✓ / No / 8X
3	5	3	5
✓ / ✗	✓ / ✗	✓ / ✗	✓ / ✗
6 / 4 / 2	6 / 4 / 2	6 / 2 / 4	6 / 2 / 4
✓	✓	✓	✓
✓	✓	✓	✓
✓	✓	✓	✓
3.00	2.50	2.17	2.50
✓ / ✓	✓ / ✓	On CD-ROM	On CD-ROM
✓	✓	✓	✓
29.27	107.70 *	83.23 *	119.70 *
38.87	227.67 *	245.67*	196.12 *
21.4	21.6	24.5	25.3
20.2	21.3	22.9	24.3
91	88	78	74
1,428	7,092 *	4,796 (6,308*)	6,500 *
1,946	3,362	2,410	2,631
2,933	2,075	2,961	2,948
3,387	1,837	3,656	3,761
457	2,915	1,176	2,915
2,591	3,352	3,457	3,904
7,858 / 3,246	7,823 / 3,246	8,347 / 3,439	8,321 / 3,442
18,862 / 19,294	18,885 / 19,429	20,033 / 20,583	20,017 / 20,567
1,883	2,125	2,330	2,832
1,806	2,039	2,191	2,658
19,049	20,151	26,215	26,265
41.89	57.15	59.14	63.28
14.20	13.00	14.87	14.80
56.09	70.15	74.01	78.08
14.02	24.62	17.01	20.02
4,000	2,850	4,350	3,900

* Were obtained by using a Geforce4 MX-440 AGP 8X Display card

Mercury's KT400FDSX and Krypton C18G-400 clawed at each other, returning nearly equal scores. In *Quake III Arena*, the C18G-400 took a good lead, leaving both the boards behind. In the 3D Mark2001 SE test, it was Mercury's turn to outshine the others by logging the best scores for a board in this category. Overall, it was Mercury's KT400FDSX that got the better of these two Krypton boards in the gaming arena by a very small margin.

In the real-world tests, it was no surprise that the Krypton motherboards outdid the Mercury's KT400FDSX by a significant margin. In the synthetic tests, especially the PCMark 2004 test, Krypton's NF2400U came out tops, beating both the boards with an outstanding performance.

Boards With a Purpose

Along with the usual breed of desktop boards, we also got two boards—P4SCA and P4SSA—that were targeted at niche markets. Both these boards were sent to us by SuperMicro. The P4SCA is based on Intel's 875P desktop chipset, while the P4SSA model is based on Intel's E7205 workstation chipset.

The P4SCA is targeted as a desktop chipset, but was not part



For Rs 16,000, you get backward compatibility for ISA components with an Intel 875P chipset

of the comparison as it lacks an AGP slot. Since the integrated RageXL display adapter is quite old, we decided not to run our graphic benchmarks on this board.

The USP of this board is the presence of three ISA (Industry Standard Architecture) slots—the long black slots that were seen on motherboards of the 486 and Pentium era. Apparently, this is the only board in the market to integrate ISA slots on Intel's latest flagship chipset-875P. The presence of ISA slot is strange, since nearly all expansion cards today are PCI based. When we asked why this was so, we were informed that the ISA standard is still prevalent in the telecom sector. So the P4SCA is targeted at customers that require legacy compatibility with the latest technology. Apart from the missing 8x AGP slot, this board offers all other Intel 875P features—dual-channel memory, Gigabit Ethernet, eight USB ports, etc. The board performed as well as the other 875-based boards—except for the sore graphics. Priced at Rs, 16,900, it's expensive by desktop standards, but could be indispensable for the telecom sector.

The P4SSA is a workstation board, but by today's desktop standards it is outdated. It has support for 533 FSB processors, DDR 266 memory modules in dual-channel with ECC, Gigabit Ethernet, six USB 2.0 ports, AGP 8X slots, etc. All these features have been superseded by desktop boards based on 875 chipset. Hence, we recommend that you buy a board based on later chipset models. We found that it offers a decent performance—somewhere between the Intel 845 and Intel 865 chipset boards. Priced at Rs 12,800, this board is too expensive. Opt for an Intel 875PBZ motherboard instead.

Conclusion

As for the boards based on the KM266 chipset, this is the end of the world and they can take it no longer. Coming to the nForce2-based boards, things get a bit complex. With good performance, the Krypton C18G-400 offers a better price-to-performance ratio as compared to the similarly priced KM400-based ASUS A7V8X-MX, which was the next best among the onboard graphic boards. Hence, if you don't mind running your PC using onboard graphics, but do wish to keep that upgrade option around for the future, then nothing beats the Krypton C18G-400.

However, if you are ready to shell out a few extra bucks needed for a graphics card, then the best board to pair it with is the Krypton NF2400U—courtesy its superb performance. At Rs 2,850, the value for money that Mercury's KT400FDSX provides simply can't be beaten; the lack of an onboard graphics controller notwithstanding.